(vi) This process may be continued for all the components.

(vii) Shade/colour each sector with different shades/colours.

(viii) Write legends to each component.

Example 4.6

Draw a pie diagram for the following data (in hundreds) of house hold expenditure of a family.

Items	Expenditure
Food	87
Clothing	24
Recreation	11
Education	13
Rent	25
Miscellaneous	20

Also find

(i) The central angle of the sector corresponding to the expenditure incurred on Education

(ii) By how much percentage the recreation cost is less than the Rent.

Solution :

The following procedure is followed to draw a Pie diagram for a given data:

(i) Calculate the total expenditure, say, N.

(ii) Compute angles for each component food, clothing, recreation, education, rent and miscellaneous using the formula $\frac{class\ frequency}{N} \times 360$

Item	Expenditure	Angle of the circle
Food	87	$\frac{87}{180} \times 360 = 174$
Clothing	24	$\frac{24}{180} \times 360 = 48$
Recreation	11	$\frac{11}{180} \times 360 = 22$
Education	13	$\frac{13}{180} \times 360 = 26$
Rent	25	$\frac{25}{180} \times 360 = 50$